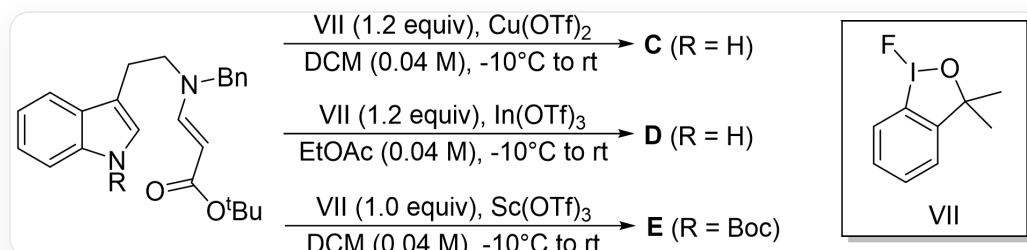


Question

Using β -aminocrotonate as a substrate, different alkaloid skeletons can be constructed under the catalysis of different Lewis acids:



Substrate is O=C(OC(C)(C)C)/C=C/N(CC1=CC=CC=C1)CCC2=CN(C3=CC=CC=C32)[R], reaction conditions are: 1.2 equivalents of VII (structure is CC1(C)OC(F)C2=CC=CC=C21), LA, solvent, -10 degrees Celsius gradually increased to room temperature, substrate concentration is 0.04 M. When LA=Cu(OTf)2, R=H, solvent=DCM, the product is **C**; when LA=In(OTf)3, R=H, solvent=EtOAc, the product is **D**; when LA=Sc(OTf)3, R=Boc, solvent=DCM, the product is **E**

It is known that **C** has a six-fused-five-spiro-six structure, **D** has a six-fused-five-fused-seven structure, and the molecular formula of **E** is C_{25}H_{25}N_2O_4. **C**, **D**, and **E** do not contain fluorine, and their formation all involves a key intermediate $\mathbf{X}^{2+}$ that does not contain iodine. The C–I bond has never been formed during the reaction. Which of the following statements are correct:

- C** has 10 $\text{sp}^2\text{C} - \text{H}$ bonds
- If benzene rings are not considered, **D** has 2 carbon-carbon double bonds
- E** has 4 rings
- If R=H, the key intermediate $\mathbf{X}^{2+}$ has 16 $\text{sp}^3\text{C} - \text{H}$ bonds

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 1,2

G. 1,3

H. 1,4

I. 2,3

J. 2,4

K. 3,4

L. 1,2,3

M. 1,2,4

N. 1,3,4

O. 2,3,4

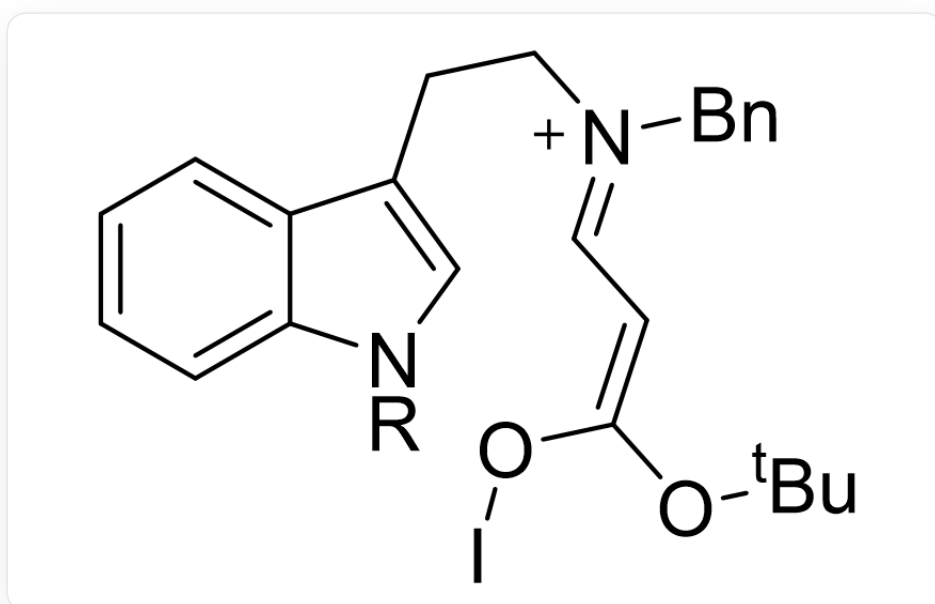
P. 1,2,3,4

Answer

Correct Answer: **J**

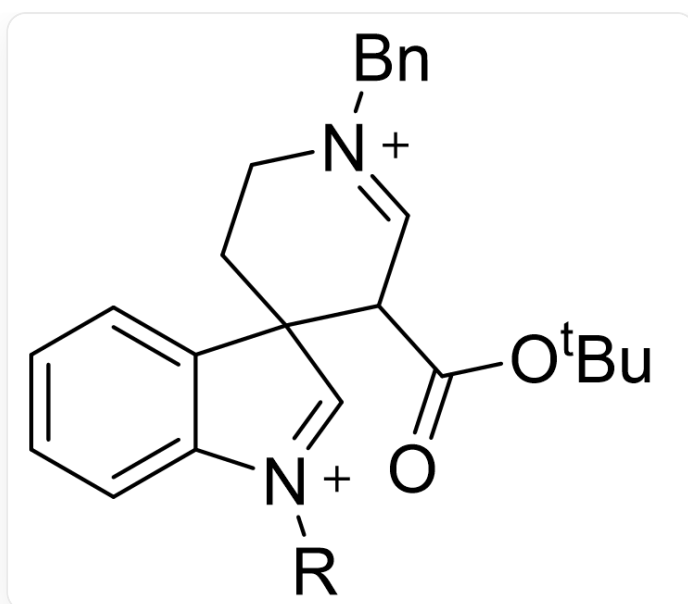
Detailed Explanation

VII is an oxidant. According to the information in the question, oxidation can only be completed by forming I – O bonds. Therefore, the following intermediate is formed first (where "I" represents the iodine atom and its connected groups):



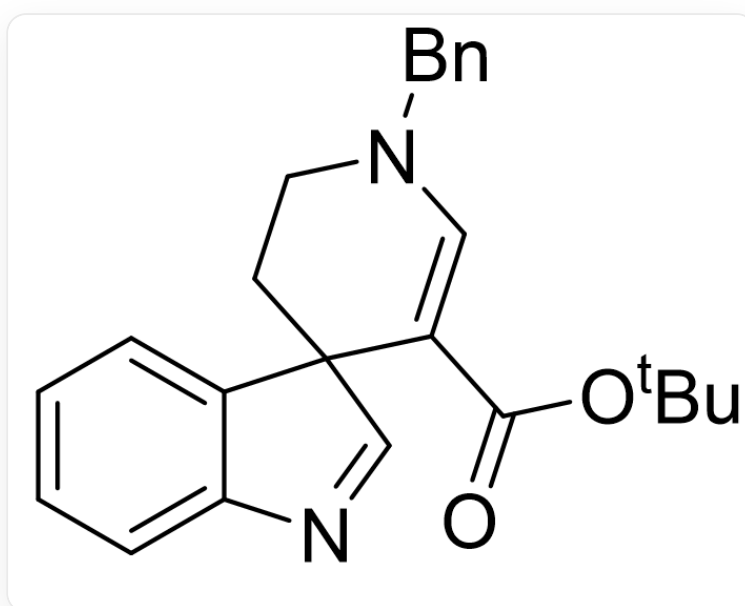
IO/C(OC(C)(C)C)=C\C=[N+](CC1=CC=CC=C1)\CCC2=CN(C3=CC=CC=C32)[R]

C has a hexacyclopenta[6,5]spiro[6] structure. Only the C3 carbon of indole can undergo nucleophilic attack, then the key intermediate $\mathbf{X}^{2+}$ is obtained (it is not yet certain that it is $\mathbf{X}^{2+}$ at this time; this is written this way for convenience):



O=C(OC(C)(C)C)C(C=[N+](CC1=CC=CC=C1)CC2)C32C=[N+](R)C4=CC=CC=C43

After losing 2 protons, **C** can be obtained:



O=C(OC(C)(C)C)C1=CN(CC2=CC=CC=C2)CCC31C=NC4=CC=CC=C43

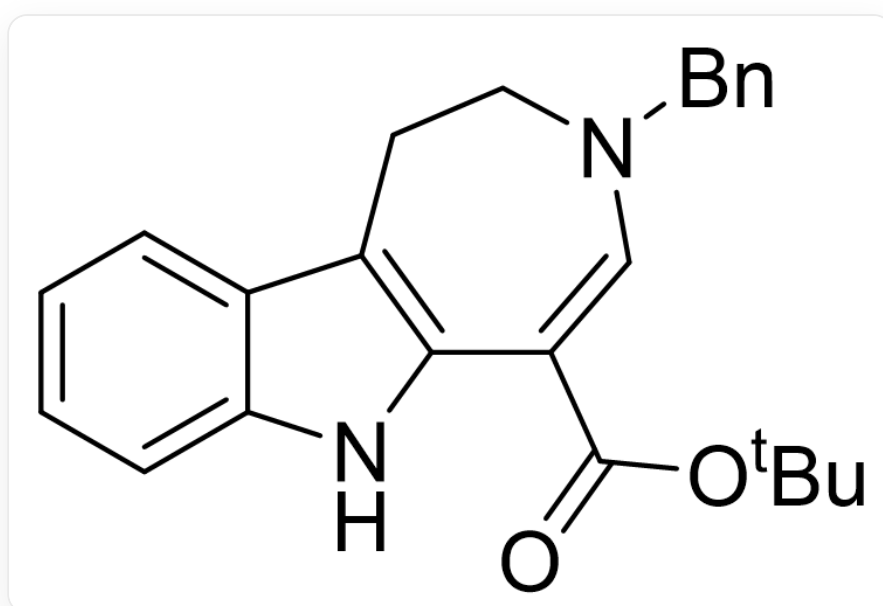
CHECKPOINT

1 PTS

C is O=C(OC(C)(C)C)C1=CN(CC2=CC=CC=C2)CCC31C=NC4=CC=CC=C43

There are $5+4+1+1=11$ $\text{sp}^2\text{C}-\text{H}$ bonds in **C**, statement 1 is incorrect.

To obtain the hexacyclopentahepta[6,5,7] **D**, alkyl migration is required, and after losing 2 protons, **D** can be obtained:



O=C(OC(C)(C)C)C1=CN(CC2=CC=CC=C2)CCC3=C1NC4=CC=CC=C43

CHECKPOINT

1 PTS

D is O=C(OC(C)(C)C)C1=CN(CC2=CC=CC=C2)CCC3=C1NC4=CC=CC=C43

Ignoring the benzene ring, **D** has 2 carbon-carbon double bonds, statement 2 is correct.

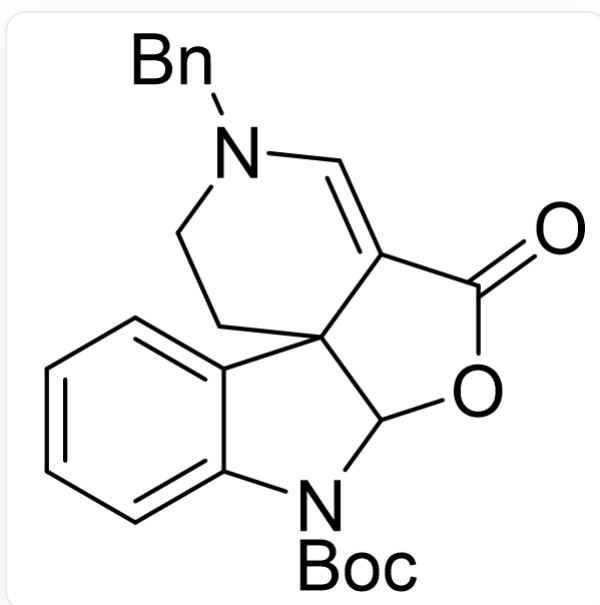
It can be basically determined that $\mathbf{X}^{2+}$ is O=C(OC(C)(C)C)C(C=[N+])(CC1=CC=CC=C1)CC2C=[N+](R)C4=CC=CC=C4. Considering the benzyl and tert-butyl groups, there are a total of $2+2+2+1+9=16$ $\text{sp}^3\text{C}-\text{H}$ bonds, statement 4 is correct.

CHECKPOINT

1 PTS

$\mathbf{X}^{2+}$ is O=C(OC(C)(C)C)C(C=[N+])(CC1=CC=CC=C1)CC2C=[N+](R)C4=CC=CC=C4

From the molecular formula of $\mathbf{E}$, it can be seen that the tert-butyl group connected to the ester group is removed. Then, the carbonyl oxygen will undergo nucleophilic attack, that is, the carboxyl oxygen in $\mathbf{X}^{2+}$ attacks the imine cation to form a five-membered ring, and after losing a proton, $\mathbf{E}$ is obtained:



O=C1OC2N(C3=CC=CC=C3C24C1=CN(CC5=CC=CC=C5)CC4)C(OC(C)(C)C)=O

CHECKPOINT

1 PTS

$\mathbf{E}$ is O=C1OC2N(C3=CC=CC=C3C24C1=CN(CC5=CC=CC=C5)CC4)C(OC(C)(C)C)=O

E contains 5 rings, statement 3 is incorrect.

Statements 2 and 4 are correct, choose J.